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Foot-and-Mouth Disease: A Comprehensive Overview

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Abstract

This comprehensive review explores various aspects of Foot-and-Mouth Disease (FMD), a globally significant disease of animals that affect animals with cloven hooves. The causative agent of this disease is a virus that belongs to family Picornaviradae and genus Apthovirus. Each year 2.35 billion shots of FMD vaccination are given to cattle worldwide. The transmission modes are multifaceted i.e. FMD can spreads by direct contact, contaminated fomites, airborne particles, and international trade, underscoring the need for a detail understanding to devise effective control strategies. The virus principally affects the epithelial cells of organs such as mouth, oropharynx and foot. Clinical signs and symptoms such as detailing fever, vesicular lesions, lameness, and reduced milk production are seen. The diagnosis methods involve the use of laboratory methods such as virus neutralization tests, ELISAs, and RT-PCR for accurate identification and serotyping. As the treatment of FMD is only supportive so the main focus should be to eradicate or limit the disease. Vaccination campaigns are often targeted, focusing on high-risk areas or regions with a history of FMD outbreaks. Disinfection with sodium carbonate solution (4%) and other products should be done frequently to decrease the viral existence in the environment.

Keywords: foot and mouth disease; economy; vaccination; diagnosis; control; prevention

Introduction

Animal health is severely affected by FMD worldwide. It's known as most important transboundary animal disease (Diaz-San Segundo et al., 2017). It influences both domesticated and wild animals with cloven hooves. Among the worst affected are the buffalo, pigs, cattle, sheep, goats, and other animals that are commonly kept by humans. The cause of this disease is a virus in nature and is classified under family Picornaviradae and belonging to the group known as Apthovirus. The FMD virus is an RNA virus, non-enveloped and the capsid geometry is icosahedral in nature and is made of single strands. This capsid harbor the viral genome which consists of an RNA molecule with a positive polarity. This virus has baseline organized molecular structures carrying 60 copies of structural proteins (You SH et al., 2019). The size of the viral particle ranges from 22 to 30. In terms of output, export bans, costs of immunizations and reduction in tourists' arrivals to the contaminated zones, epidemic of this infectious disease has associated risks of tremendous economic losses for nations. For example, each year 2.35 billion shots of FMD vaccination are given to cattle worldwide, and the anticipated

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Regions	Serotypes
Asia	A, O, ASIA 1
Africa	A, O, C, SAT 1, SAT 2, SAT 3
Europe	A, O, C
South America	A, O, C
Oceania	FMD Free
North and Central America	FMD Free
Caribbean	FMD Free

Table 1: *Geographical distribution of different FMD Serotypes*
Sourced from "A review on foot and mouth disease in dairy animals, etiology, pathogenesis and clinical finding Azeem et al., 2020, Pure and Applied Biology (PAB), 9(1), 821-832.

total cost at peak cost (US$9 per dose) is around US$20.7 billion (Kardjadi and Mustafa 2018). The FMD virus has high variability in its genetic makeup, especially in the areas that code for surface proteins, which leads to the development of several serotypes and subtypes. The creation of vaccines is hampered by this diversity, which also emphasizes the necessity for ongoing study and surveillance to comprehend and control the virus's dynamic nature. Generally speaking, China, India, and Africa see the greatest economic effects from FMD. In Africa specifically, the disease's economic impact is underestimated, typical vaccine regimens are too expensive, control of the illness is not yet prioritized, and its epidemiology is unclear, despite of its huge sector in the production sector that is US$2.32 billion (Casey et al., 2018). Due to the prolonged virus shedding from affected animals and the restoration of some species of animals, controlling FMD requires significant resources utilization (Firestone et al.,2018). Because there are presently seven identified serotypes of FMD circulating globally, the disease is viewed as seven immunologically separate diseases when viewed through the lens of epidemiology and disease control (Lycett et al., 2019). Because of this, animals that develop immunity against single variant of FMDV are still susceptible to other strains and serotypes of the virus and the degree of immunity against different variants depends on antigenic profiles (Anna and James 2015). Furthermore, the fact that established protocols call for an extensive slaughter of diseased and perhaps "in contact" animals when epidemics occur in FMD-free zones makes FMD an issue of concern for animal welfare (Brito et al., 2017). The necessity to isolate and cull infected individuals, makes the financial impact of an epidemic on a nation even more grave, in spite of the extent of natural mortality from FMD; in other words, a finding of FMD could result in the culling the whole exposed population.

Epidemiology:

The incidence of FMD worldwide and its effects on animals are the two main characteristics of its epidemiology. Widespread in Africa, Asia, and Middle East, FMD continuously threatens vulnerable people. FMD is endemic in many countries in Africa, southern Asia and the Middle East (Jamal and Belsham 2018). The virus comes in seven different serotypes, each of which has several subtypes, adding to the complexity of the epidemiology of FMD. Certain serotypes are typically limited by geography (see Table 1). For example, the SAT strains are primarily found in sub-Saharan Africa, though SAT 2 FMDV has been found in Egypt for a number of years (Brito et al., 2017). Similarly, the Asia-1 a serotype, as its name implies, is normally found in Asia, and is spread till Greece. Once widely distributed, serotype C FMDV may already be extinct in the field because it hasn't been recorded anywhere since 2004 (Knight-Jones et al., 2016). In Pakistan for FMD seroprevalence, 1,478 small ruminants were examined. According to reports, the seroprevalence was 22.8% (Farooq et al., 2018). 9.83% of 2511 serum samples collected from animals were tested out as positive in Khyber Pakhtunkhwa province of Pakistan (Khan et al., 2016). In Saudi Arabia, there have also been reports of an outbreak at sheep farms that led to newborn deaths and abortions (100). FMDV RNA was detected in milk samples by analysis. In 5.7% (42/732) of the samples of milk, it was found (Armson et al., 2020). The seroprevalence of FMD in non-vaccinated animals in Hail was reported to be 17.5% (Mahmoud et al., 2021).

In Somalia, Foot and mouth disease has not been observed among pastoralists in the Lower Shebelle region of Somalia. Based on A KAP study, it was found that many of the respondents were aware of the economic impact and financial implications to livestock owners brought by foot and mouth disease (Barre Abdirahman, et al., 2024).

A total of 1080 cattle, 840 sheep and 2080 goat sera samples were tested with ELISA for FMDV 3-ABC NSP in another study in somalia. The seroprevalence of FMD in cattle was 18.52% (200/1080) and was significantly higher (20%) in the summer season (Elnaker, Y. F. et al., 2012). Cattle are highly susceptible, there is increased contact during seasonal movements and summer is favorable to viral transmission. Factors contributing are low vaccination coverage, poor biosecurity practices, and retention of virus in environment.

According to World Organization for Animal Health, among 77% of the world's cattle population this has been spread. Nations that do not have a vaccination against FMD are nonetheless constantly at risk of incursion. The two regions with the highest expenses are Africa and Eurasia, which contribute 50% and 33% of the total costs, respectively. FMD is the sickness for which WOAH has created an official list of nations free of the illness, which can be acknowledged as being free of the illness overall or in specific areas and sections

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Transmission:

Direct Contact:

A prime mean of transmission is direct contact among infected and susceptible animals (Fukai et al., 2020). The virus is present at high level in the oral and nasal fluids, as well as in the milk, urine, and feces of infected animals. Close proximity, such as in shared grazing areas or crowded housing conditions, facilitates the transfer of the virus through contact with these bodily fluids. Infected animals shed the virus even before clinical signs appear, making it challenging to detect and control the spread during the early stages of infection.

Fomites:

Indirectly, FMDV can survive and spread when spread in the environment it is able to remain viable for long durations of time in conditions where the temperature is within 50°C, and the relative humidity is more than 55%, while, pH of 7 is optimal for the virus (Colenutt et al., 2020). Vectors can spread the virus to new areas by shedding their loaded feces and other body fluids. Due to movement and contact with the affected animals or products, the farmers, veterinarians, and other infected individuals become the agents of spreading the FMD from one place to another (Guyver-Flectcher et al., 2022).

Airborne transmission:

It is another important factor in the spread of FMD since movement is not easily restricted within a given region. The virus can be transmitted through droplets in respiratory secretions and can be aerosolized over a fairly long distance. This mode of transmission is most applicable in places with increased animal contact where respiratory moistened secretion having the virus can be spread in an airborne manner (Lai et al., 2022). Various factors such as the wind and the environmental conditions help determine the survival of the virus in air such that it can spread beyond fences and barriers across long distances. The initial model of transmission includes the transmission of virus droplets through the respiratory system from a direct interaction with infected animals in their acute inflammatory stage (Paton et al., 2018).

Trade:

The transport of animals and other livestock and international trade is also another cause of FMD spread all over the world. A virus can be hailed via infected animal movement or contaminated products across borders (Shanafelt et al., 2017) Missing or ineffective quarantine, and gaps in border measures may allow the spread of the virus and the introduction of the virus into FMD-free areas with subsequent outbreaks and threats to the livestock industry and economy.

Pathophysiology:

FMDV life-cycle involves a sequence of events after exposure that prominently involves the interaction of the virus with the host. It gains access to the body and invade different parts of the body that are most suitable for virus replication including the mouth, udder, and heart, foot, and oro-pharynx (Admassu et al., 2015). The virus is known to infect the epithelial cells in oral and nasal cavities, and also skin of the feet. This virus is able to infect its host by docking to requisite receptors and thus undergo the cycle of viral reproduction. This process takes place within the host cell's cytoplasm and results in the development of new virus particles (Arzt et al., 2017).

These areas subsequently get infected, forming vesicular lesions in the process. The emergence of vesicles is thought to stem from the breakdown of infected cells so that viral particles can be released and spread throughout the neighboring cells. Due to alterations seen on oral examination characterized by the presence of vesicular lesions, the patient excessively salivates, slobbers, and is reluctant to feed. These symptoms increase speed of the spread of the virus, because the virus is excreted by sick animals in their regular secretion-like saliva.

Besides the oral cavity, the disease also affects the feet, which may ulcerate and make the animal lame and interfere with co-ordination to shuffling. The lameness resulting from the viral infection also helps the virus to spread since infected animals may rub against other healthy individuals during movement. The agent can also enter and multiply in different organs, including the lungs, spleen, and lymph nodes

After the infection's early phase, a significant percentage of animals can continue to have an infection for a long time, with minute viral load (Stenfeldt et al., 2016). It is unclear how important the carrier state is epidemiologically. A study suggests that buffaloes in carrier phase can spread the virus to cattle, but it's been not feasible to prove through experiments that animal in carrier stage can spread the virus to healthy animals (Tekleghiorghis et al., 2016). Moreover, naïve cattle were efficiently infected by directly transferring oropharyngeal fluid from carrier animals to their oropharynx (Arzt et al., 2018). As a result, there is a risk associated with the oropharyngeal fluid of carrier animals. Because of this, there is a significant motivation to minimize the risk of an outbreak, even in cases when the likelihood of carrier animal transferring the disease is very rare, due to the substantial economic consequences involved.

Clinical Signs and Symptoms of FMD:

The disease commonly starts its course with a sudden onset of fever, a systemic response to the viral infection. This is due to the distinctive development of vesicular lesions, which is a characterizing feature of FMD. These fluid-filled blisters mainly occur in the oral cavity and involve specific structures like the tongue papillae and lips or dental pad (Saguii et al., 2019). The existence of these painful lesions causes drooling, which can even be observed in the form of visible saliva coming from the patient's mouth. Not only oral lesions are presented in FMD, but also cutaneous vesicular lesions on the feet, where they are often most intensively located in the coronary bands and between the fingers. This may lead to lameness and what can be referred to as a 'shuffling' gait since the animals are trying not to put their feet on the ground.

Infectious FMD can lead to lower milk productivity in infected dairy cattle as a result of the disease, sometimes sharply reduced (Raina et al., 2023). Demonstrated through the fact that animals with oral lesions as well as potential lesions in the teats will not give efficient milk yields due to the discomfort. Thus, another loss which is also an economic effect of the disease among the farmers who rear both large and small dairy animals is the reduction in milk production.

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Weight loss is one of the clinical manifestations of FMD (Baluka, 2016). Therefore, it is possible to notice a visible deterioration in body condition due to a decrease in feed intake caused by infection, changes in carbohydrate metabolism due to the systemic effects of the virus, and pain sensations from oral and foot lesions.

Diagnosis:

The identification of strain of the FMD virus causing outbreaks of disease and timely, sensitive, and specific laboratory diagnosis are crucial to stop spread of FMD and prevent potential disastrous economic impacts which could result from an outbreak. Clinical diagnosis depends on symptoms like elevated temperature level, formation of vesicles on the oral surfaces, at area of nose, coronary bands and spaces between digits. The main method used to diagnose FMD is to find evidence of FMDV particles or the viral genome in clinical materials such as saliva, milk, semen, tongue and foot epithelium, etc. Confirmatory diagnosis is obtained by virus neutralization test and ELISA detection of intact virus particles (Sharma Et al., 2015). The present techniques are explained as follows:

Virus Neutralization Test:

VNT plays key role in certifying animals and animal products for importation or exportation, is recently set as the "gold standard" for tracing antibodies formed against structural components of FMDV (Stear and M.J 2005). If we compare VNT to other serological tests, VNTs are greater susceptible to changes because they employ different types of cell lines and cell. In comparison other drawbacks are that it takes more time and is very susceptible to contaminations so extra care along with a well specialized and equipped lab is needed.

Enzyme Linked Immunosorbent Assay:

The modern ELISA is a modified form of radioimmunoassay methods in which an antigen is either directly or indirectly immobilized on a solid phase to capture a targeted antibody. The antibody is then reported by conjugation of enzyme with secondary antibody, which will produce signals when its corresponding substrate is present (Wong et al., 2020).

Reverse Transcription Polymerase Chain Reaction:

This procedure involves the extraction of viral RNA from materials such as vesicular or oral fluid, after that the RNA undergo process of reverse transcription to form a complementary DNA (Khan et al., 2021). The cDNA is amplified by polymerase chain reaction (PCR) with certain primers. Precise quantification is possible with real-time monitoring. Identification of the FMD virus's serotype and subtype is aided by the detection of amplified DNA. This method is best for a precise diagnosis and it also saves time, hence effective in outbreak management.

Blood Profile:

Patients with FMD see specific alterations in their blood profile. The concentration of IL-4 and IL-10 was decreased in the cows suffering from FMD, while IL-1, TNF-alpha, IFN-γ, IL-6, were significantly high (El-Karim et al., 2022). In another study, interferon and major histocompatibility complexes decrease significantly in case of an infection (Li et al., 2021).

Control and Prevention:

Using killed vaccines for large scale vaccination in most cases is successful but provides temporary protection that is serotype- and sometimes strain-specific (Diaz-San et al., 2017). Among different approaches, vaccination remains one of the most pivotal tools in the framework of FMD prevention and control. Vaccination campaigns have always been specific in that they may target certain focal areas or areas where there have been previous occurrences of FMD (Singh et al., 2019).

Surveillance for FMD is crucial in the planning and implementation of a proper check and control mechanism. That is why the identification of the virus at an early stage lets to make some urgent actions and start certain measures of isolation (Halasa et al., 2015). Flawless surveillance lets authorities detect affected animals, euthanize/quarantine them swiftly, thus reducing the likelihood of disease spread.

Measures involving strict cleaning and disinfection, especially in contact points like the live-stock marketing facilities and transportation channels also help to reduce the spread of FMD. In Japan, sodium carbonate solution (4%) and several other products are often used to inactivate FMDV (Fukai et al., 2021). Some general measures taken to adopt biosecurity include restricted animal trafficking, isolation provisions, and greater cleanliness. Isolation slows potential spread of the infected animals and contaminated articles to other areas not previously affected.

Since this virus is easily transboundary in nature, international cooperation is vital in both the prevention and control of FMD. The status, outbreak incidences and control measures of FMD are well reported by countries to organizations like the World Organization for Animal Health. There are OIE Standards for controlling FMD that has brought about unity in tackling the disease across the world. More research on new adjuvants and how they may be incorporated into FMD vaccinations can enhance both immunogenicity and the possible lifespan of the protection (Kamel et al., 2019).

Conclusion

This review brings forward multiple angles of FMD pointing to the intricate interplay between virus, host and environment. FMD remains a major problem in global agriculture, and its implications remain today not only limited to the economic but have social and geopolitical characteristics. Nevertheless, vaccination continues to be essential in operational FMD control measures, however improved diagnostic tools, antiviral chemotherapies, and molecular approaches for surveillance form the basis of efficient control and prevention of FMD. This outlook coupled with the ability of the virus to move from wildlife through domestic animals to humans or in the formation of new viral strains serves to justify the multi-sectoral One Health approach. Such diseases call for collaborative efforts as well as sharing of information across the borders. Moreover, technologies and advances in farming methods that can be applied to prevent the breakdown of barriers and prevent FMD can also help in its control. In light of these challenges being faced with FMD, it is crucial to have an adequate appreciation of virology, epidemiology and socio-economic impact surrounding the disease to develop feasible and sustainable methods of reducing the effects of FMD in the livestock industries worldwide

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